國立高雄科技大學 商務資訊系
2026創新資訊與行動應用競賽 系統概述

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Introduction

Berth Allocation and Quay Crane Assignment Problem(BACAP) is about the synchronized optimization of:

- spatial berth allocation (**where** vessels dock along the quay)
- temporal resource allocation (**how many and which** quay cranes service each vessel over time).

This project is about:

An environment simulating Berth Allocation and Quay Crane Assignment Problem(BACAP) for Reinforcement Learning(RL) model training;

A GUI-enabled platform for port topology design, which allows user to design port topology through GUI, rather than coding

Scope Limitation

The purview, undertakings, and deliverables of this project **shall be strictly, narrowly, and exclusively confined to, and limited to, the virtual construction of the environment and the construction of the web-based platform.**

All forms, manifestations, and attributes of visualization—specifically including, without limitation, **visual clarity, graphical legibility, aesthetic refinement, and beautification—are hereby categorically omitted, excluded, and excised from the project.** No obligation, warranty, or deliverable pertaining to visualization of any kind shall attach hereto.

All metrics, measurements, optimizations, evaluations, enhancements, and assessments relating to model performance are strictly omitted from the project. The project team bears no responsibility, liability, or undertaking whatsoever regarding the operational efficacy, predictive accuracy, or optimization of any RL model.

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How is this creative

Normally, the BACAP is solved by mathematical optimization, which makes assumptions about the "form" of the solution. We do not follow the traditional path ; instead, this project developed the environment of BACAP for it to be solved by a Reinforcement Learning (RL) algorithm, thereby relaxing the assumptions on the solution.

Another part of this project involves a web-like platform that allows user to design port topology using GUI, rather than writing code

Brief Introduction for the Functionalities

Please note what we are doing here, we do not aim to build a complete RL model, as it requires too much computational resources. This project is about developing the environment simulating BACAP to enable RL training

Features of the System

Stochastic Environmental Dynamics

Environmental volatility is modeled using a 5-dimensional Ornstein-Uhlenbeck stochastic differential equation (SDE) coupled via a defined correlation matrix Σ. The drift term pulls variables toward equilibrium (μ), while the diffusion term incorporates Wiener process increments (dW) transformed by the Cholesky factor of the covariance matrix:

- $\mathbf{X}_{t+\Delta t} = \mathbf{X}_t + \theta(\mu - \mathbf{X}_t)\Delta t + \mathbf{L}d\mathbf{W} + 0.1(d\mathbf{W}^2 - \Delta t)$
- $\boldsymbol{\theta} = [0.15,0.10,0.08,0.12,0.10]^T$ represents the mean-reversion rates.
- $\boldsymbol{\mu} = [6.5,0.9,14.0,0.15,0.5]^T$ represents the long-term equilibrium mean values.
- $\boldsymbol{\sigma} = [1.4,0.18,0.5,0.04,0.05]^T$ represents the volatility coefficients.
- L is the Cholesky factor matrix derived from the covariance matrix $\Sigma = \text{diag}(\boldsymbol{\sigma})C\text{diag}(\boldsymbol{\sigma})$, where C is the environmental cross-correlation matrix.
- $d\mathbf{W}_t$ is standard Brownian motion increment vector distributed as $\mathcal{N}(0,dt \cdot I)$.

To account for higher-order stochastic integration accuracy, the discrete update incorporates a Milstein-type correction term:

$\mathbf{X}_{t+\Delta t} = \mathbf{X}_t + \boldsymbol{\theta} \odot (\boldsymbol{\mu} - \mathbf{X}_t)\Delta t + d\mathbf{W}_t + 0.1\left((d\mathbf{W}_t)^2 - \Delta t\right)$
where $\Delta t = \mathrm{dt} = 0.25$ hours. Physical variables are subsequently clamped to non-negative or bounded physical domains (e.g., visibility clipped to [0.1,25.0]).

This formulation ensures mean-reverting physical behaviors for wind, waves, visibility, precipitation, and water currents, capturing temporal persistence and cross-variable correlations (e.g., wind-wave coupling).

Why Milstein?

Limitations of the Euler-Maruyama Scheme

Consider a general multi-dimensional Itô SDE governed by drift vector $a(X_t)$ and diffusion matrix $b(X_t)$:

$dX_t = a(X_t)dt + b(X_t)dW_t$
The standard Euler-Maruyama discretization approximates this as:

$X_{t+\Delta t} = X_t + a(X_t)\Delta t + b(X_t)\Delta W_t$
where $\Delta W_t \sim \mathcal{N}(0, \Delta t \cdot I)$. While Euler-Maruyama has a weak convergence order of $\mathcal{O}(\Delta t)$, its **strong convergence order** (measuring pathwise trajectory error $\mathbb{E}[\,|X_t - X_t^{\text{approx}}|\,] \to 0$]) is only $\mathcal{O}(\sqrt{\Delta t})$(**see** Oxford's Lecture). In reinforcement learning environments where agents learn policies dependent on precise temporal trajectories of environmental stress, first-order path discretization errors can lead to systemic policy bias or failure to capture rare, high-consequence environmental transitions.

The Milstein Expansion and Itô-Taylor Derivation

The Milstein method is derived from the stochastic Taylor expansion (Itô-Taylor expansion) by retaining higher-order derivative terms of the diffusion coefficient. For a scalar or diagonal noise formulation, the Milstein scheme takes the form:

$$X_{t+\Delta t} = X_t + a(X_t)\Delta t + b(X_t)\Delta W_t + \frac{1}{2}b(X_t)b'(X_t)(\Delta W_t^2 - \Delta t)$$
where $b'(X_t) = \frac{\partial b}{\partial X}$. This elevates the strong convergence order from $\mathcal{O}(\sqrt{\Delta t})$ to $\mathcal{O}(\Delta t)$ **(see** Oxford's Lecture**, as above)**.

NOTE: because our time step $\Delta T = 0.25 < 1$, convergence $\mathcal{O}(\Delta t)$ is better than $\mathcal{O}(\sqrt{\Delta T})$

Instrumental Value for Quayside CMDP Training

From an instrumentalist engineering viewpoint, the inclusion of this term provides three distinct operational advantages:

- **Variance Reduction in Extreme States**: By accounting for the stochastic curvature term ($\Delta W^2 - dt$), the simulation avoids underestimating the variance of environmental volatility spikes (waves, wind gusts).
- **Prevention of Policy Overfitting**: RL agents trained under Euler-maruyama paths often exploit deterministic or smoother noise realizations. The Milstein correction injects rigorous second-order noise characteristics, forcing the PPO agent's Lagrangian constraint mechanism to handle realistic stochastic turbulence.
- **Compatibility with $\Delta t = 0.25$ Hours**: At a coarse discretization time step of 15 minutes ($\Delta t = 0.25$), first-order truncation errors compound rapidly. The Milstein correction stabilizes sample paths without requiring sub-minute integration loops that would degrade training throughput.

Why Cholesky?

Cholesky decomposition is deployed to factorize the environmental covariance matrix $\Sigma \in \mathbb{R}^{5\times5}$ into a lower triangular matrix L, enabling the transformation of orthogonal standard normal white noise into cross-correlated Wiener increments for the Ornstein-Uhlenbeck (OU) stochastic differential equations.

Exact Preservation of Physical Covariance Structure:

Maritime terminal simulations fail if environmental variables are modeled as independent random walks. In reality, wind speed and wave height are strongly coupled (assumed to be positively correlated at $r = 0.75$ in the environment's correlation matrix). By applying the Cholesky factor L to independent raw increments $d\mathbf{W}_{\text{raw}} \sim \mathcal{N}(0, dt \cdot I)$:

$$d\mathbf{W} = L\, d\mathbf{W}_{\text{raw}}$$

the resulting stochastic increments strictly preserve the target covariance:

$$\mathbb{E}[d\mathbf{W} d\mathbf{W}^T] = L\, \mathbb{E}[d\mathbf{W}_{\text{raw}} d\mathbf{W}_{\text{raw}}^T]\, L^T = L(dt \cdot I)L^T = (LL^T)dt = \Sigma\, dt$$

This guarantees that the reinforcement learning agent trains against physically valid, synchronized environmental shocks (e.g., a sudden wind storm correctly and simultaneously driving up wave heights according to historical covariance), maximizing real-world applicability and policy robustness.

Computational Optimality and Minimal Overhead:

Alternative approaches for generating correlated noise—such as eigenvalue (spectral) decomposition or general Singular Value Decomposition (SVD)—require computing full orthogonal eigenvectors and singular values. For a 5×5 matrix, Cholesky decomposition executes in roughly $\frac{1}{3}n^3$ operations, requiring fewer than 50 floating-point operations. This ensures that environmental stepping adds zero perceptible latency to the simulation loop.

Mathematical Uniqueness and Numerical Stability:

For any symmetric positive-definite covariance matrix Σ, the Cholesky decomposition yields a *unique* lower triangular matrix L with strictly positive diagonal entries. This eliminates stochastic initialization ambiguities, ensuring deterministic mathematical behavior under fixed random seeds while avoiding the phase or magnitude distortions associated with heuristic noise generation techniques.

Harmonic Tidal Model

Water level variation (tide height H_{tide}) as a function of time t (in hours) is modeled by superimposing principal harmonic tidal constituents:

$$H_{\text{tide}}(t) = A_{M_2}\cos(\omega_{M_2}t) + A_{S_2}\cos(\omega_{S_2}t) + A_{M_4}\cos(\omega_{M_4}t)$$

where the angular frequencies ω and amplitudes A are defined as:

- M_2 **Principal Lunar Semidiurnal**: $A_{M_2} = 2.8\text{m}, \omega_{M_2} = \frac{2\pi}{12.4206}$ rad/hr
- S_2 **Principal Solar Semidiurnal**: $A_{S_2} = 1.2\text{m}, \omega_{S_2} = \frac{2\pi}{12.0000}$ rad/hr
- M_4 **Shallow Water Terdiurnal**: $A_{M_4} = 0.3\text{m}, \omega_{M_4} = \frac{4\pi}{12.4206}$ rad/hr

Crane Kinematics and Collision Avoidance

Each crane $i \in \{1, \ldots, N_c\}$ maintains position p_i, velocity v_i, and target position p_i^{target}. The kinematic equations update position and velocity subject to acceleration limits ($a_{\max} = 15.0$ m/h^2), velocity limits ($v_{\max} = 45.0$ m/h), and mechanical friction ($\gamma_f = 0.02$):

$$a_i = \text{clip}\left(v_i^{\text{target}} - v_i, -a_{\max}, a_{\max}\right) - \gamma_f v_i$$
$$v_i^{(t+\Delta t)} = v_i^{(t)} + a_i \Delta t$$
$$p_i^{(t+\Delta t)} = p_i^{(t)} + v_i^{(t+\Delta t)} \Delta t$$

To prevent physical collisions between adjacent cranes, an iterative projection loop enforces minimum inter-crane separation distance $d_{\min} = \text{crane_distance} = 30.0$ m:

$$\forall i, j \quad \text{s.t.} \quad p_i < p_j \implies p_j \geq p_i + d_{\min}$$

Vessel Hydrostatics, Ballast, and Stability Mechanics

For each berthed vessel, cargo unloading proceeds at a rate influenced by environmental efficiency multipliers (wind, rain, visibility, and congestion):

$$\eta_{\text{env}} = \underbrace{\max\left(0,1 - 0.06\max(0, v_{\text{wind}} - 10)\right)}_{\text{Wind Factor}} \times \underbrace{\max\left(0.1,1 - 0.15P_{\text{rain}}\right)}_{\text{Rain Factor}} \times \underbrace{\min\left(1,\frac{\text{Vis}}{5}\right)}_{\text{Visibility Factor}} \times \underbrace{\max\left(0.5,1 - 0.05N_{\text{berthed}}\right)}_{\text{Congestion Factor}}$$

The operational handling rate R_{op} determines cargo mass reduction and vertical center of gravity (KG) evolution:

$$\Delta M_{\text{cargo}} = R_{\text{op}} \times 15.0 \times \Delta t$$
$$KG^{(t+\Delta t)} = KG^{(t)} + 0.001 \times R_{\text{op}} \times \Delta t$$

Ballast water adjustments driven by action rate $a_{\text{ballast}} \in [-1,1]$ update ballast mass M_{ballast} and free surface correction (FSC):

$$M_{\text{ballast}}^{(t+\Delta t)} = \text{clip}\left(M_{\text{ballast}}^{(t)} + a_{\text{ballast}} \times 50.0 \times \Delta t, 500.0, 5000.0\right)$$
$$\text{FSC} = 0.2 \times \left(\frac{M_{\text{ballast}}}{3000.0}\right)$$

Transverse metacentric height (GM) and heeling angle (heel) are calculated using foundational ship stability equations:

$$\nabla = \frac{M_{\text{vessel}}}{1.025}$$
$$\text{BM} = 4.5 \times \left(\frac{L_{\text{vessel}}}{300.0}\right), \quad KM = 5.0 + \text{BM}$$
$$GM = KM - KG - \text{FSC}$$
$$\text{Heeling Moment} = (\Delta M_{\text{ballast}} \times 12.0) + (\Delta M_{\text{cargo}} \times 4.0)$$
$$\text{Heel} = \text{clip}\left(\frac{\text{Heeling Moment}}{\max(0.5, GM)}, -15.0, 15.0\right)$$

Constraint violations occur if $GM < 0.5$m or $|\text{heel}| > 5.0°$, incurring Lagrangian cost penalties in the CMDP formulation.

Reward Function

Mathematically, the total reward R_t computed at time step t is defined as:

$$R_t = \left(\alpha_1 R_{\text{queue}} + \alpha_2 R_{\text{berth}} + R_{\text{shaping}}\right)\Delta t + \alpha_3 R_{\text{completion}}$$

Each component is detailed below:

1. Queue Congestion Penalty (R_{queue})

Penalizes the ratio of active vessels in the queue relative to the maximum queue capacity:

$$R_{\text{queue}} = -\left(\frac{N_{\text{queue}}}{Q_{\text{capacity}}}\right)^2 - 0.05 \max\left(0.0, \frac{N_{\text{queue}}}{Q_{\text{capacity}}} - 0.5\right)$$

where N_{queue} is the current number of vessels waiting in the queue, and Q_{capacity} is the maximum queue capacity (set to 15).

2. Berth Allocation Reward (R_{berth})

Evaluates the immediate success or failure of placing a vessel into a berth, scaled by the vessel's priority weight:

$$R_{\text{berth}} = \left(2.0 \cdot \mathbb{I}(\text{berth_success}) - 0.5 \cdot \mathbb{I}(\text{berth_failure})\right) \cdot \left(\frac{w_{\text{priority}}}{3.0}\right)$$

where $\mathbb{I}(\cdot)$ is an indicator function, and $w_{\text{priority}} \in [1.0, 3.0]$ represents the vessel's priority weight.

3. Potential-Based Reward Shaping (R_{shaping})

Provides intermediate credit assignment based on the reduction of remaining cargo move balances across all active vessels (in queue and berthed):

$$R_{\text{shaping}} = 0.1\left(\gamma \Phi(s_{t+1}) - \Phi(s_t)\right)$$

where the potential function $\Phi(s)$ is defined as the negative sum of normalized remaining move balances:

$$\Phi(s) = -\sum_{v \in \text{active}} \frac{\text{move_balance}_v}{\max\left(1.0, \text{move_sum}_v\right)}$$

4. Vessel Completion Reward ($R_{\text{completion}}$)

Rewards successful completion of vessel operations, granting a high bonus for on-time delivery ($\tau_v = 0.0$) and a linear penalty for tardiness ($\tau_v > 0.0$):

$$R_{\text{completion}} = \sum_{v \in \text{finished}} \left(\frac{w_{\text{priority},v}}{3.0} \right) \cdot \begin{cases} 100.0, & \text{if } \tau_v = 0.0 \\ -0.05 \cdot \tau_v, & \text{if } \tau_v > 0.0 \end{cases}$$

where $\tau_v = \max(0.0, t_{\text{current}} - t_{\text{due}})$ is the tardiness duration of vessel v.

Reward Shaping

Define the potential before the transition as:

$$\text{potential_before} = - \sum_{v \in \mathcal{V}_{\text{active}}} \frac{\text{move_balance}_v}{\max\left(1.0, \text{move_sum}_v\right)}$$

After the environment transitions and updates the vessel workloads, the potential after the transition is computed as:

$$\text{potential_after} = - \sum_{v \in \mathcal{V}_{\text{active, after}}} \frac{\text{move_balance}_v}{\max\left(1.0, \text{move_sum}_v\right)}$$

The reward shaping term is then incorporated as:

$$\text{reward_shaping} = 0.1 \times \left(\gamma \cdot \text{potential_after} - \text{potential_before}\right)$$

This structure corresponds directly to **Potential-Based Reward Shaping (PBRS)**, which modifies the original reward function $R(s, a, s')$ to accelerate reinforcement learning convergence without altering the set of optimal policies.

系統開發工具與技術:python and its module:gymnasium, for env. HTML, CSS, Javascript for port design system
系統使用對象:anyone who is interested in solving BACAP using RL
系統使用環境:should be general, as long as the related modules are installed correctly
結語: we developed the system and proved it works
介紹短片網址：link

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